WORKED SOLUTION

STEP II 1993, Question 7

Pure • Infinite Descent Modulo 5

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q7

The key idea

The right-hand side is a multiple of 5, so working modulo 5 turns the equation into a small finite check: a square can only be 0, 1 or 4 $\pmod 5$. Once a and b are forced to be multiples of 5, the equation forces c to be one too, and dividing everything by 5 gives a *smaller* solution. A non-zero integer can't be divided by 5 for ever, so this is Fermat's method of infinite descent.

Solution

Divisibility of a and b by 5

Squaring each residue modulo 5:

$x \pmod 5$	0	1	2	3	4
$x^2 \pmod 5$	0	1	4	4	1
$2x^2 \pmod 5$	0	2	3	3	2

So $b^2 \equiv 0, 1$ or 4, and $2a^2 \equiv 0, 2$ or $3 \pmod 5$. Since $5c^2 \equiv 0 \pmod 5$, we need

$$2a^2 + b^2 \equiv 0 \pmod 5.$$

Now we list every combination:

$2a^2 + b^2$	$b^2 \equiv 0$	$b^2 \equiv 1$	$b^2 \equiv 4$
$2a^2 \equiv 0$	**0**	1	4
$2a^2 \equiv 2$	2	3	1
$2a^2 \equiv 3$	3	4	2

The only entry that is 0 is the one with $2a^2 \equiv 0$ and $b^2 \equiv 0$. From the first table, $x^2 \equiv 0 \pmod 5$ only when $x \equiv 0$ (equivalently, 5 is prime, so $5 \mid x^2$ implies $5 \mid x$). Hence $a \equiv 0$ and $b \equiv 0 \pmod 5$: both a and b are divisible by 5, as required.

The only solution is $a = b = c = 0$

Write $a = 5p$ and $b = 5q$ with p, q integers. Substituting,

$$25(2p^2 + q^2) = 5c^2 \quad \Longrightarrow \quad c^2 = 5(2p^2 + q^2).$$

So $5 \mid c^2$, and since 5 is prime, $5 \mid c$. Write $c = 5r$. Then $25r^2 = 5(2p^2 + q^2)$, that is,

$$2p^2 + q^2 = 5r^2.$$

This is the original equation again. So if (a, b, c) is a solution, then all three are divisible by 5 and $(a/5,\ b/5,\ c/5)$ is also a solution in integers.

> **Ask yourself** Why does "we can keep dividing by 5" force everything to be zero?

Apply the step repeatedly. Starting from a solution (a, b, c), after n steps we have the integer solution $(a/5^n,\ b/5^n,\ c/5^n)$, so 5^n divides each of a, b and c, for *every* $n \geq 1$.

But a non-zero integer m can only be divided by 5 a limited number of times: if $5^n \mid m$ and $m \neq 0$ then $5^n \leq |m|$, which fails once n is large. So none of a, b, c can be non-zero.

Equivalently, as a descent argument: suppose a solution existed with $c \neq 0$, and pick one with $|c|$ as small as possible and positive. Then $(a/5, b/5, c/5)$ is a solution with $0 < |c/5| < |c|$, a contradiction. So $c = 0$, and then $2a^2 + b^2 = 0$ forces $a = b = 0$, since both terms are non-negative.

Hence the only integer solution is $a = b = c = 0$.

Checks

- Run through all 25 pairs of residues (a, b) mod 5: only $(0, 0)$ makes $2a^2 + b^2 \equiv 0$.
- A computer search over all $|a|, |b| \leq 300$ finds no solution of $2a^2 + b^2 = 5c^2$ other than $a = b = c = 0$.
- Contrast with $a^2 + b^2 = 5c^2$, which *does* have non-trivial solutions such as $(1, 2, 1)$: there $1 + 4 \equiv 0 \pmod 5$, so the residue argument in the first part breaks down, exactly as it should.

Common slips

- Concluding $5 \mid a$ from $5 \mid a^2$ without a reason. It's true because 5 is prime (or by the table), but it needs saying.
- Forgetting to show that c is divisible by 5 as well. Without it, $(a/5, b/5, c/5)$ need not be a solution in integers, and the descent doesn't start.
- Stopping after "we can divide by 5 again". The proof isn't complete until you say why repeated division forces zero.

Beyond STEP

This is Fermat's *infinite descent*, the same method that proves $\sqrt{2}$ is irrational and that $x^4 + y^4 = z^2$ has no non-trivial solutions. Equations like $2a^2 + b^2 = 5c^2$ are studied systematically in number theory: Legendre's theorem gives exact conditions, in terms of quadratic residues, for when $Ax^2 + By^2 = Cz^2$ has a non-trivial integer solution. Here the obstruction is that -2 is not a square modulo 5. Davenport's *The Higher Arithmetic* covers congruences, quadratic residues and descent.

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